

Silicate Melt Architecture: Network Formers, Modifiers & Magma Rheology

Comprehensive Textbook Study Guide & Complete Solution for DGMS 2019 Exam (Question 24)

Standard References: William M. White, *Geochemistry* (2005) | John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010) | Dexter Perkins, *Mineralogy*

PART I

Theoretical Foundation & Direct Textbook Analysis

Detailed conceptual walkthrough of silicate liquid structure, bridging vs. non-bridging oxygens, cation classification, and melt viscosity

1. THE ATOMIC ARCHITECTURE OF SILICATE LIQUIDS & GLASSES

A foundational insight in modern igneous petrology is that silicate magmas are **not random, unstructured ionic fluids**. Spectroscopic studies (Raman, IR, NMR, and EXAFS) of silicate glasses—which represent supercooled liquids that preserve the structural state of the melt—reveal that liquids retain the short-range atomic architecture of the crystalline minerals that crystallize from them.

"Most, though not all, of our knowledge of the structure has come from studies of glasses rather than melts. While the thermodynamic properties of silicate liquids and their respective glasses differ, other studies have confirmed the general structural similarities of glasses and liquids. Spectral studies of glasses, which in some respects can be viewed as supercooled liquids, have revealed that silicate liquids have structures rather similar to those of silicate solids. In fact, the principal difference between silicate liquids and solids is the absence of long-range ordering in the former; short range ordering is similar. As in silicate minerals, the primary structural element of silicate liquids is the silicon tetrahedron, consisting of a silicon atom surrounded by four oxygens."

— William M. White, *Geochemistry* (2005), Chapter 4: Applications of Thermodynamics, p. 136

The primary structural unit in all silicate melts is the $[\text{SiO}_4]^{4-}$ **tetrahedron**, with a Si–O distance of $\sim 1.62 \text{ \AA}$ and O–Si–O bond angles of $\sim 109.5^\circ$. Tetrahedra link together by sharing oxygen atoms:

- **Bridging Oxygen (BO, O^0):** An oxygen shared between two tetrahedral centers ($\equiv\text{Si}-\text{O}-\text{Si}\equiv$ or $\equiv\text{Si}-\text{O}-\text{Al}\equiv$). It has a formal charge of zero with respect to the network, linking tetrahedra into continuous polymer chains, sheets, and 3D frameworks.
- **Non-Bridging Oxygen (NBO, O^-):** An oxygen bonded covalently to only one tetrahedral silicon atom ($\equiv\text{Si}-\text{O}^-$). Its extra unit of negative charge is balanced by electrostatic attraction to nearby modifier metal cations (Ca^{2+} , Mg^{2+} , Fe^{2+} , Na^+ , K^+).

- **Free Oxygen (FO, O^{2-}):** An oxygen anion not bonded to any silicon atom, present only in highly depolymerized, ultrabasic, or carbonatitic melts and coordinated entirely by metal cations.

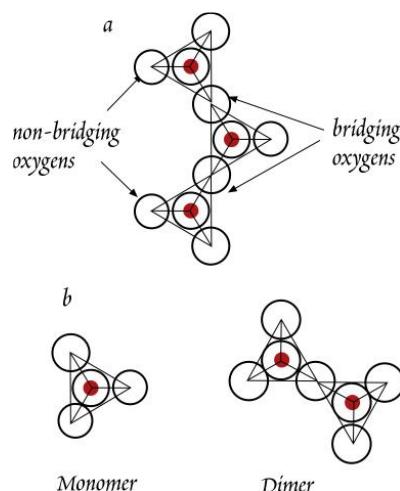


Figure 4.21 (White, *Geochemistry*, 2005, p. 136): Silicate structures in melts. (a) Short-range silicate structures in melts resemble those in solids. Individual tetrahedra may be linked by bridging oxygens shared between two silicon atoms, or terminated by non-bridging oxygens. (b) Structural units in silicate melts include isolated monomers ($[SiO_4]^{4-}$, with zero bridging oxygens) and dimers ($[Si_2O_7]^{6-}$, where only 1 of 4 oxygens in each tetrahedron is bridging).

2. CATION CATEGORIZATION: NETWORK FORMERS VS. NETWORK MODIFIERS

How a dissolved cation behaves in a silicate melt depends fundamentally on its **ionic potential (Z/r , valence divided by ionic radius)** and its ability to fit into tetrahedral 4-fold coordination with oxygen, as governed by **Pauling's First Rule (Radius Ratio Rule)**.

*"The degree to which the silica tetrahedra are linked, or polymerized, in silicate liquids affects chemical and physical properties. The degree of polymerization in turn depends on other cations present. **These may be divided into two groups, network formers and network modifiers.** Relatively small, highly charged cations such as Al^{3+} and Fe^{3+} (more rarely, Ti^{4+} , P^{3+} , B^{3+} as well) often substitute for silicon in tetrahedral sites and, along with Si, are termed network formers. **The other common cations of natural silicate liquids, Ca^{2+} , Mg^{2+} , K^+ , Na^+ , and H^+ , are network modifiers. These ions cannot substitute for silicon in tetrahedra and their positive charges can only be balanced by non-bridging oxygens. Addition of these ions disrupts the linkages between silica tetrahedra. Thus as silicate melts become richer in these network modifiers they become progressively depolymerized.**"*

— William M. White, *Geochemistry* (2005), Chapter 4: Applications of Thermodynamics, p. 136

Crystallographic Reason: Why Ca^{2+} and Mg^{2+} Cannot Form Networks

According to Pauling's Rules, the coordination number (CN) of a cation surrounded by oxygen anions ($r_{\text{O}^{2-}} \approx 1.40 \text{ \AA}$) is dictated by the radius ratio $r_{\text{cation}} / r_{\text{oxygen}}$:

- **Tetrahedral (CN = 4) stability window:** $0.225 \leq r/r_{\text{O}} < 0.414$.

Si^{4+} ($r = 0.26 \text{ \AA}$, $r/r_{\text{O}} = 0.19$) and Al^{3+} ($r = 0.39 \text{ \AA}$, $r/r_{\text{O}} = 0.28$) fit easily into tetrahedral sites.

- **Octahedral (CN = 6) stability window:** $0.414 \leq r/r_{\text{O}} < 0.732$.

Mg^{2+} ($r = 0.72 \text{ \AA}$, $r/r_{\text{O}} = 0.514$) is far too large for a tetrahedron and requires 6-fold octahedral coordination.

- **Cubic / Antiprismatic (CN = 8) stability window:** $0.732 \leq r/r_{\text{O}} < 1.000$.

Ca^{2+} ($r = 1.00 \text{ \AA}$, $r/r_{\text{O}} = 0.714$ to 0.75) requires 6- to 8-fold coordination.

Because Ca^{2+} and Mg^{2+} cannot occupy tetrahedral sites, they sit in the open interstitial spaces between tetrahedra. To achieve local charge balance, they pull electrons from adjacent oxygens, rupturing the $\equiv\text{Si}-\text{O}-\text{Si}\equiv$ bridging bonds:

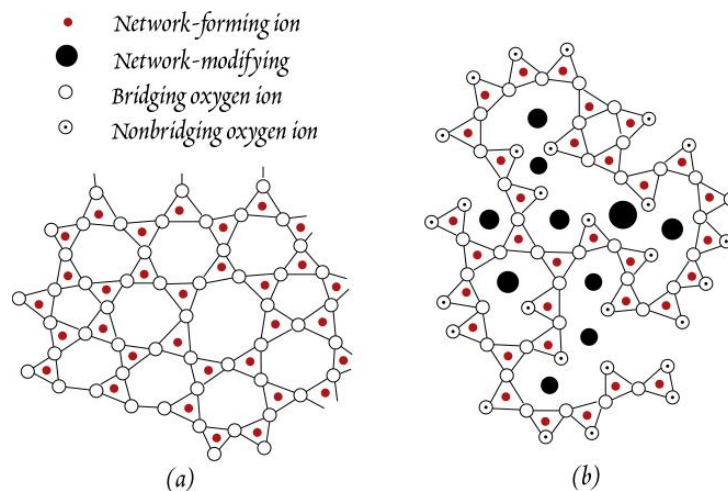


Figure 4.22 (White, *Geochemistry*, 2005, p. 136): (a) Structure of pure silica glass (liquid), consisting of a fully polymerized, continuous 3D network where all oxygens are bridging (BO). (b) A silica-rich glass with additional component ions: network-modifying cations (solid black circles, e.g., Ca^{2+} , Mg^{2+}) enter the structure, severing bridging bonds and creating non-bridging oxygens (open circles, NBO), thus disrupting the continuous polymer network.

3. QUANTIFYING DEPOLYMERIZATION: THE NBO/T PARAMETER & TRACE ELEMENTS

To parameterize the structural state of magmas in quantitative petrology, scientists use the **NBO/T ratio (Non-Bridging Oxygens per Tetrahedral Cation)**:

$$\text{NBO/T} = (2\text{O} - 4\text{T}) / \text{T}$$

where **O** is total oxygen atoms per formula unit and **T** is total tetrahedral cations ($\text{Si}^{4+} + \text{Al}^{3+} + \text{Fe}^{3+} + \text{P}^{5+}$).

- **Quartz / Pure SiO_2 Melt:** $\text{O} = 2, \text{T} = 1 \Rightarrow \text{NBO/T} = (4 - 4)/1 = \mathbf{0.0}$ (Completely polymerized framework).
- **Feldspar Melt ($\text{NaAlSi}_3\text{O}_8$):** $\text{O} = 8, \text{T} = 4 \Rightarrow \text{NBO/T} = (16 - 16)/4 = \mathbf{0.0}$ (Fully polymerized; Na^+ charge-balances Al^{3+}).
- **Pyroxene Melt ($\text{CaMgSi}_2\text{O}_6$):** $\text{O} = 6, \text{T} = 2 \Rightarrow \text{NBO/T} = (12 - 8)/2 = \mathbf{2.0}$ (Infinite single chains).
- **Olivine Melt (Mg_2SiO_4):** $\text{O} = 4, \text{T} = 1 \Rightarrow \text{NBO/T} = (8 - 4)/1 = \mathbf{4.0}$ (Completely depolymerized isolated monomers).

"The ratio of non-bridging oxygens to tetrahedral cations (Si^{4+} and Al^{3+}), written NBO/T, provides a measure of the degree of melt depolymerization. Kohn and Schofield (1994) carried out a series of experiments demonstrating the effect of melt polymerization on olivine/melt partitioning of Zn and Mn... the Zn partition coefficient decreases from about 4.5 in highly polymerized melts (low NBO/T) to about 0.8 in depolymerized melts (high NBO/T)."

— William M. White, *Geochemistry* (2005), Chapter 7: Trace Elements, p. 278

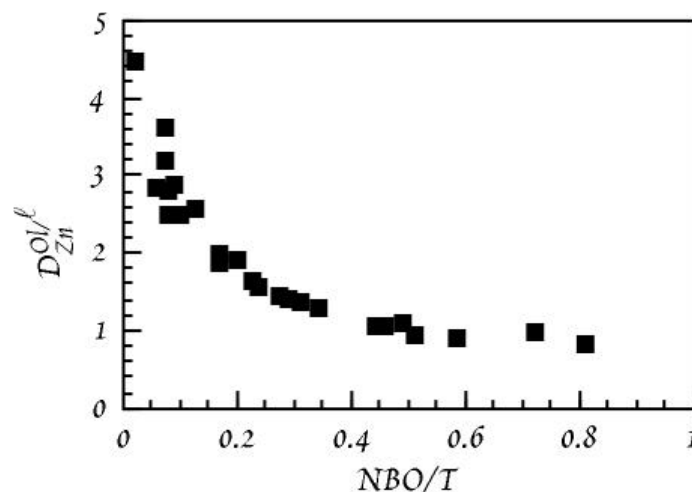


Figure 7.13 (White, *Geochemistry*, 2005, p. 278): Variation of the zinc olivine/liquid partition coefficient ($D_{\text{Zn}}^{\text{ol/l}}$) as a function of the number of non-bridging oxygens per tetrahedral cation (NBO/T) in experiments by Kohn and Schofield (1994). In depolymerized melts (high NBO/T, rich in Ca^{2+} and Mg^{2+}), abundant octahedral sites exist in the liquid, lowering the partition coefficient.

Confirmation in Thermodynamic Activity Models

In the two-lattice thermodynamic melt model of Nielsen & Dungan (1983) presented in White (2005, p. 278, Eqs. 7.35–7.36), network modifier oxides (ΣNM) and network former oxides (ΣNF) are separated mathematically:

$$\begin{aligned}\Sigma\text{NM} &= X_{\text{AlO}1.5} + X_{\text{CaO}} + X_{\text{MgO}} + X_{\text{FeO}} + X_{\text{P}_2\text{O}_5} + X_{\text{TiO}_2} + X_{\text{MnO}} + X_{\text{CrO}1.5} - \\ &\quad X_{\text{NaO}0.5} - X_{\text{KO}0.5} \\ \Sigma\text{NF} &= X_{\text{SiO}_2} + X_{\text{NaO}0.5} + X_{\text{KO}0.5}\end{aligned}$$

Note that CaO and MgO are explicitly summed as network modifiers (ΣNM), providing mathematical textbook confirmation.

4. MELT RHEOLOGY: VISCOSITY, DENSITY & THE ROLE OF VOLATILES (H_2O VS. CO_2)

Melt structure is the single most important factor controlling the physical behavior of erupting magmas. High polymerization creates an interconnected network of strong Si–O–Si and Si–O–Al bonds that strongly resists viscous shear flow.

"Because melts approximate the structures of the corresponding minerals, basaltic melts tend to have more isolated Si-O complexes, with no bridging oxygens (as in olivine) and partially bridged -Si-O-Si-O- chains (as in pyroxene), than fully bridged three-dimensional -Si-O-Al-O- networks (as in plagioclase). Because they yield much more feldspar and quartz, rhyolitic melts tend to be dominated by three-dimensional -Si-O-Al-O- networks. The more polymerized -Si-O-Al-O- networks create strong interconnected bonds throughout the rhyolitic melt, and thus higher viscosity. Greater viscosity thus generally correlates with higher silica content."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010), Chapter 3, p. 61

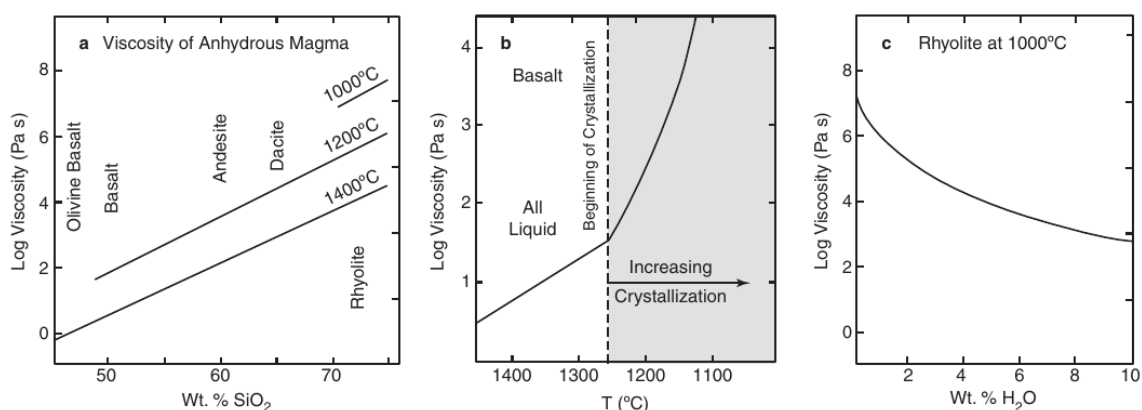


Figure 1 (Winter, *Principles of Igneous and Metamorphic Petrology*, 2010, p. 62): (a) Calculated viscosities of anhydrous silicate liquids at 1 atmosphere pressure (Bottinga & Weill, 1972; Hess, 1989): Rhyolite viscosity is $\sim 10^4$ Pa·s at 1400°C and skyrockets to 10^7 Pa·s at 1000°C, whereas depolymerized

basalt is ~ 1 Pa·s. (b) Abrupt increase in viscosity of basalt as it crystallizes. (c) Dramatic reduction in rhyolite viscosity at 1000°C as dissolved H_2O depolymerizes the melt network (Shaw, 1965).

Contrasting Volatiles: H_2O (Depolymerizer) vs. CO_2 (Polymerizer)

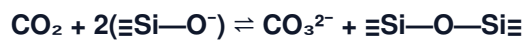
Winter (2010, Chapter 5, pp. 145–146) details a profound contrast in how dissolved volatile gases interact with silicate melt structures:

- **Water (H_2O):** Dissolves by attacking bridging oxygens:



H_2O behaves as a powerful **network modifier**. Adding just 2 wt% H_2O to a rhyolitic melt drops its viscosity by two full orders of magnitude (from 10^7 to 10^5 Pa·s; Figure 1c above).

- **Carbon Dioxide (CO_2):** In marked contrast, CO_2 does not attack bridging bonds. Instead, in mafic melts rich in Ca^{2+} and Mg^{2+} , CO_2 reacts with two non-bridging oxygens to form carbonate complexes:



CO_2 actually consumes non-bridging oxygens and produces bridging oxygens, increasing the degree of polymerization of the melt! Therefore, CO_2 raises melt viscosity and dissolves preferentially in depolymerized, Ca-Mg-rich mafic melts.

PART II

DGMS 2019 Examination: Question & Worked Solution

Rigorous option-by-option critical analysis and definitive textbook-grounded answer

ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

Q.24. In the atomic structure of silicate melts, Ca^{2+} and Mg^{2+} ions are categorized as

- (a) Network modifiers
- (b) Network formers
- (c) Bridging atoms
- (d) Non-bridging atoms

Correct Answer: (a) Network modifiers



Ca^{2+} and Mg^{2+} are large, divalent cations that cannot enter 4-fold tetrahedral coordination with oxygen. They occupy interstitial positions and disrupt the polymerized framework by converting bridging oxygens into non-bridging oxygens.

OPTION-BY-OPTION CRITICAL ANALYSIS

✓ (a) Network modifiers

CORRECT

Why it is correct: As explicitly established in White (2005, p. 136) and Winter (2010, p. 61), cations in silicate liquids are classified based on whether they build or break the tetrahedral polymer network. Because of their large ionic radii ($\text{Mg}^{2+} \approx 0.72 \text{ \AA}$, $\text{Ca}^{2+} \approx 1.00 \text{ \AA}$) and low charge (+2), Ca^{2+} and Mg^{2+} cannot enter tetrahedral sites. Instead, they occupy interstitial 6- to 8-fold coordinated sites where their charges are satisfied by attracting non-bridging oxygens. **This disrupts the $\equiv\text{Si}-\text{O}-\text{Si}\equiv$ linkages, depolymerizing the melt, lowering its viscosity, and increasing its density.** Hence, they are universally classified as network modifiers.

X (b) Network formers

INCORRECT

Why it is incorrect: Network formers are small, highly charged cations (Si^{4+} , Al^{3+} , Fe^{3+} , and minor Ti^{4+} , P^{5+}) with ionic potentials $Z/r \geq 6$ that can enter 4-fold tetrahedral coordination ($[\text{TO}_4]$). Pauling's radius ratio rule requires $0.225 \leq r/r_{\text{O}} < 0.414$ for tetrahedral coordination. With r/r_{O} values of 0.514 (Mg^{2+}) and 0.714 (Ca^{2+}), these cations are crystallographically excluded from tetrahedral network sites.

X (c) Bridging atoms

INCORRECT

Why it is incorrect: In silicate melt nomenclature, the term "bridging" applies exclusively to **oxygen anions (O^0)** that are shared between two adjacent $[\text{SiO}_4]$ tetrahedra. Ca^{2+} and Mg^{2+} are positively charged metal counterions, not bridging atoms.

X (d) Non-bridging atoms

INCORRECT

Why it is incorrect: "Non-bridging" also strictly describes an **oxygen species (O^-)** bonded to a single silicon atom ($\equiv\text{Si}-\text{O}^-$). While Ca^{2+} and Mg^{2+} associate electrostatically with non-bridging oxygens, the cations themselves are categorized functionally as **network modifiers**.

PART III

Synthesis Reference Tables & High-Yield Practice Questions

Quick-reference diagnostic tables and 8 curated exam questions for DGMS, UPSC GSI, and IIT-JAM

SYNTHESIS TABLES: CATION CLASSIFICATION & MAGMA PROPERTIES

Table 1: Comprehensive Classification of Cations in Silicate Melts

Category	Cations	Coordination (CN)	Structural Action	Effect on Polymerization
Network Formers	Si ⁴⁺ , Al ³⁺ , Fe ³⁺ (Ti ⁴⁺ , P ⁵⁺ , B ³⁺)	4 (Tetrahedral)	Build 3D -Si-O-Si- / -Si-O-Al- framework	Increases polymerization & viscosity
Network Modifiers	Ca ²⁺ , Mg ²⁺ , Fe ²⁺ , Mn ²⁺ , Na ⁺ , K ⁺ , H ⁺	6 to 8 (Octahedral / Interstitial)	Rupture bridging bonds; create NBOs	Depolymerizes melt; lowers viscosity
Charge Balancers	Na ⁺ , K ⁺ (coupled with Al ³⁺)	Near [AlO ₄] ⁻ tetrahedra	Maintains neutrality: Na ⁺ + Al ³⁺ ↔ Si ⁴⁺	Allows Al ³⁺ to act as network former
Amphoteric / Intermediate	Fe ³⁺ , Al ³⁺ , Ti ⁴⁺	4 or 6 (Depends on alkalinity/P/T)	Former in alkaline; modifier if excess	Variable depending on peralkalinity

Table 2: Comparison Across the Natural Magma Spectrum

Magma Type	SiO ₂ (wt%)	Key Modifiers	NBO/T	Dominant Units	Viscosity (Pa·s)	Density (g/cm ³)
Rhyolite / Granite	70–76%	Very Low (Alkalis balance Al)	0.02–0.15	3D Framework (Tectosilicate-like)	10 ⁵ – 10 ⁸	~2.20 – 2.35
Andesite / Diorite	57–63%	Moderate	0.30–0.50	Framework + Sheets/Chains	10 ² – 10 ⁴	~2.45 – 2.60
Basalt / Gabbro	45–52%	High (Ca ²⁺ , Mg ²⁺ , Fe ²⁺ rich)	0.60–0.95	Short Chains + Monomers	1 – 10 ²	~2.65 – 2.85
Komatiite / Ultramafic	40–45%	Extreme (MgO up to 30%)	1.20–1.80	Monomers ([SiO ₄] ⁴⁻) dominant	0.01 – 0.1	~2.85 – 3.05

Table 3: The Three Oxygen Species in Silicate Melts

Oxygen Species	Symbol	Coordination & Bonding	Formal Charge	Dominant In
Bridging Oxygen	O ⁰ (BO)	Bonded to 2 tetrahedral cations (Si–O–Si, Si–O–Al)	0 (Neutral in network)	Rhyolites, Dacites, Granites (Polymerized)
Non-Bridging Oxygen	O ⁻ (NBO)	Bonded to 1 tetrahedral cation + 1 modifier (Si–O ⁻ ... M ²⁺)	-1 (Local ionic bond)	Basalts, Andesites, Alkaline melts
Free Oxygen	O ²⁻ (FO)	Unbonded to Si; bonded only to modifier cations (M ²⁺ ... O ²⁻)	-2 (Purely ionic)	Ultra-depolymerized melts (Carbonatite, Komatiite)

PROBABLE HIGH-YIELD EXAM QUESTIONS (UPSC GSI, DGMS, IIT-JAM & GATE)

Q1. In silicate melts, Al³⁺ acts as a network former only when:

- (a) Water content is greater than 5 wt%
- (b) It is charge-balanced by an equimolar proportion of alkali (Na⁺, K⁺) or alkaline earth (Ca²⁺) cations
- (c) Melt temperature exceeds 1200°C
- (d) Fe³⁺ is absent from the melt

→ **Correct Answer: (b)**

Explanation: Al³⁺ substitutes for Si⁴⁺ in tetrahedral sites ([AlO₄]⁵⁻). Because it has a charge of +3 instead of +4, an adjacent cation (such as Na⁺, K⁺, or ½Ca²⁺) is required to maintain local electrical neutrality. Without charge balancing, excess Al³⁺ enters 6-fold coordination as a network modifier.

Q2. For a hypothetical melt having pure forsterite stoichiometry (Mg₂SiO₄), the value of NBO/T is:

- (a) 0.0 (b) 1.0 (c) 2.0 (d) 4.0

→ **Correct Answer: (d) 4.0**

Explanation: Total oxygens O = 4, tetrahedral cations T = 1 (Si). NBO/T = (2O – 4T)/T = (2 × 4 – 4 × 1)/1 = (8 – 4)/1 = 4.0. In forsterite melt, all tetrahedra are isolated monomers [SiO₄]⁴⁻ with zero bridging oxygens.

Q3. Which of the following cations has the highest tendency to act as a network modifier in silicate melts?

- (a) Si⁴⁺ (b) Ti⁴⁺ (c) P⁵⁺ (d) Ca²⁺

→ **Correct Answer: (d) Ca²⁺**

Explanation: Ca²⁺ has low charge (+2) and large radius (1.00 Å), giving it a very low ionic potential (Z/r = 2.0) and 8-fold coordination preference. Si⁴⁺, Ti⁴⁺, and P⁵⁺ are all high field-strength cations that act as network formers.

Q4. The addition of MgO or CaO to an anhydrous rhyolitic melt causes:

- (a) Increase in viscosity and decrease in density
- (b) Decrease in viscosity and increase in density
- (c) Increase in both viscosity and density
- (d) Decrease in both viscosity and density

→ **Correct Answer: (b) Decrease in viscosity and increase in density**

Explanation: As stated in White (2005, p. 136): Network modifiers rupture bridging bonds, depolymerizing the melt into shorter segments (lowering viscosity) and collapsing the open tetrahedral framework into a closer-packed structure (increasing density).

Q5. How does dissolved CO₂ affect the polymerization of mafic silicate melts compared to H₂O?

- (a) Both CO₂ and H₂O depolymerize the melt equally
- (b) H₂O depolymerizes the melt, whereas CO₂ reacts with non-bridging oxygens to increase polymerization
- (c) CO₂ depolymerizes the melt, whereas H₂O polymerizes it
- (d) Neither CO₂ nor H₂O has any effect on melt structure

→ **Correct Answer: (b)**

Explanation: Winter (2010, p. 146) highlights that CO₂ reacts with two non-bridging oxygens to produce carbonate (CO₃²⁻) and a bridging oxygen (Si–O–Si): $\text{CO}_2 + 2(\text{Si}-\text{O}^-) \rightleftharpoons \text{CO}_3^{2-} + \text{Si}-\text{O}-\text{Si}$. This creates bridging bonds and increases polymerization, directly contrasting with H₂O which breaks bridging bonds.

Q6. Divalent transition metals (e.g., Zn²⁺, Mn²⁺) have olivine/melt partition coefficients (D) that:

- (a) Increase dramatically as the melt becomes more depolymerized (higher NBO/T)
- (b) Decrease significantly as the melt becomes more depolymerized (higher NBO/T)
- (c) Are entirely independent of melt polymerization
- (d) Drop to zero in basaltic liquids

→ **Correct Answer: (b) Decrease significantly as the melt becomes more depolymerized**

Explanation: Kohn & Schofield (1994) and White (2005, p. 278) showed that D_{Zn} drops from ~4.5 in highly polymerized melts (low NBO/T) to ~0.8 in depolymerized melts (high NBO/T), because depolymerized melts provide abundant energetically favorable 6-fold octahedral sites that retain Zn²⁺ in the liquid.

Q7. Which mineral group has an atomic structure most analogous to a silicate melt dominated by dimer units [Si₂O₇]⁶⁻?

- (a) Nesosilicates
- (b) Sorosilicates
- (c) Inosilicates
- (d) Phyllosilicates

→ **Correct Answer: (b) Sorosilicates**

Explanation: Dimers consist of two [SiO₄] tetrahedra sharing exactly one bridging oxygen ([Si₂O₇]⁶⁻, NBO/T = 3.0), which is the defining structural hallmark of sorosilicates (e.g., epidote, hemimorphite).

Q8. In the 2-lattice melt solution model of Nielsen & Dungan (1983) and White (2005, p. 278), the sum of network modifiers (ΣNM) explicitly includes:

- (a) SiO₂ and NaO_{0.5}
- (b) CaO and MgO
- (c) AlO_{1.5} and SiO₂
- (d) Only H₂O

→ **Correct Answer: (b) CaO and MgO**

Explanation: In thermodynamic melt activity modeling, mole fractions of CaO, MgO, FeO, and MnO are grouped in the network-modifier component (ΣNM), whereas SiO_2 is the primary component in the network-former term (ΣNF).

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Source Exam: DGMS 2019 (Junior Scientific Officer), Question 24 · Subject: Petrology & Geochemistry

Textbook Authorities: William M. White, *Geochemistry* (2005), pp. 136–138, 278–280 | John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010), pp. 61–62, 145–146 | Dexter Perkins, *Mineralogy*